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Betting Against Beta

Replicating Frazzini & Pedersen (2014)

Frazzini, A. & Pedersen, L. H. (2014). Betting Against Beta. Journal of Financial Economics, 111(1), 1–25.

Topics covered in this lecture:

- The BAB anomaly: economic intuition and theoretical framework
- Frazzini–Pedersen beta estimator: overlapping returns & Vasicek shrinkage
- Beta-sorted decile portfolios and the empirical Security Market Line
- BAB factor construction: rank-weighting and market-neutral scaling
- Backtesting, robustness checks, and honest replication assessment

Lecture Roadmap

1	**What Is Betting Against Beta?** *Economic intuition, leverage constraints, and theoretical predictions*	~15 min
2	**Building the Signal: Estimating Beta** *Frazzini–Pedersen estimator, overlapping returns, Vasicek shrinkage*	~15 min
3	**Statistical Validation** *Beta-sorted decile portfolios, the empirical Security Market Line*	~15 min
4	**Constructing the BAB Factor** *Rank-weighting, beta-neutral scaling, portfolio mechanics*	~10 min
5	**Backtesting** *Performance analysis, drawdowns, comparison to benchmarks*	~15 min
6	**Comparing to the Paper** *Replication assessment: what matches, what differs, and why*	~10 min
7	**Robustness & Limitations** *Sub-period analysis, sector-neutrality, honest assessment*	~10 min
8	**Conclusions & Discussion** *Key takeaways, limitations, practical implications*	~5 min

1

What Is Betting Against Beta?

Economic intuition, leverage constraints, and the theoretical framework

The Crowded Elevator Problem

The Core Insight

- Rational investors optimize by combining the risk-free asset with the tangency portfolio — adjusting risk via leverage.
- Many investors (mutual funds, pensions, retail) CANNOT use leverage. To get more return they buy riskier high-beta stocks instead.
- This crowding bids UP high-beta stock prices and DEPRESSES their expected returns relative to CAPM.
- Low-beta stocks (utilities, consumer staples) are under-demanded and earn higher risk-adjusted returns.

The Central Hypothesis

- High-beta stocks are overpriced relative to CAPM because leverage-constrained investors bid them up.
- A portfolio that is LONG leveraged low-beta stocks and SHORT de-leveraged high-beta stocks — the BAB factor — earns positive risk-adjusted returns.

Key Theoretical Predictions to Test

- Proposition 1: A stock's CAPM alpha is $\alpha = \psi^{-}(1 - \beta)$, which DECREASES in beta → alpha is positive for low-β stocks, negative for high-β stocks.
- Proposition 2: The BAB factor — long leveraged low-beta, short de-leveraged high-beta — has a POSITIVE expected return.
- Proposition 4: Beta dispersion (spread) compresses during funding crises, reducing BAB expected returns in those periods.

Theoretical Framework: The Constrained SML

Standard CAPM — Security Market Line

$$E[r^s] = r^f + \beta^s \cdot (E[r^M] - r^f)$$

Perfect markets: every investor can leverage freely. The SML is steep and linear.

Frazzini–Pedersen (2014) — Constrained Equilibrium

$$E[r^s] = r^f + \bar{\psi}_t + \beta^s \cdot \lambda_t$$

where $\bar{\psi}_t > 0$ is average tightness of funding constraints and λ_t is the market risk premium.

Low-Beta Stocks (β < 1)

- Earn MORE risk-adjusted return than CAPM predicts.
- CAPM alpha: $\alpha = \bar{\psi}(1 - \beta) > 0$ → positive.
- Under-held by constrained investors → higher expected return.

High-Beta Stocks (β > 1)

- Earn LESS risk-adjusted return than CAPM predicts.
- CAPM alpha: $\alpha = \bar{\psi}(1 - \beta) < 0$ → negative.
- Over-held by constrained investors → lower expected return.

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Building the Signal: Estimating Beta

Overlapping 3-day returns, split estimation windows, and Vasicek shrinkage

The Frazzini–Pedersen Beta Estimator

Beta Decomposition (Equation 14)

$$\hat{\beta}^s = \hat{\rho}^s \cdot (\hat{\sigma}^s / \hat{\sigma}^M)$$

Volatility (σ) and correlation (ρ) are estimated on DIFFERENT rolling windows to reduce noise.

Component	Window	Rationale
Volatility: $\hat{\sigma}^s, \hat{\sigma}^M$	**1 year (252 days)**	Volatility is fast-moving; short window captures recent regime shifts
Correlation: $\hat{\rho}^s$	**5 years (1,260 days)**	Correlation is slow-moving; long window reduces estimation noise
Returns for correlation	**3-day overlapping log-returns**	Mitigates non-synchronous trading effects (thin stocks)

Vasicek Shrinkage (Equation 15)

$$\hat{\beta}^s_{final} = 0.6 \cdot \hat{\beta}^s_{TS} + 0.4 \cdot 1$$

Bayesian shrinkage toward the cross-sectional mean of 1. Reduces estimation noise for stocks with sparse data.

Data Description and Sample Characteristics

865	16 yrs	4,024	11.3%
U.S. Stocks	Sample Period	Trading Days	SPX Ann. Return

Sample & Universe

- Period: Jan 2010 – Dec 2025 (16 years, 4,024 trading days).
- Universe: 865 U.S. large- and mid-cap stocks across all 11 GICS sectors.
- Market proxy: S&P 500 Index (SPX) as the market portfolio.
- Price data: adjusted close prices (splits and dividends accounted for).

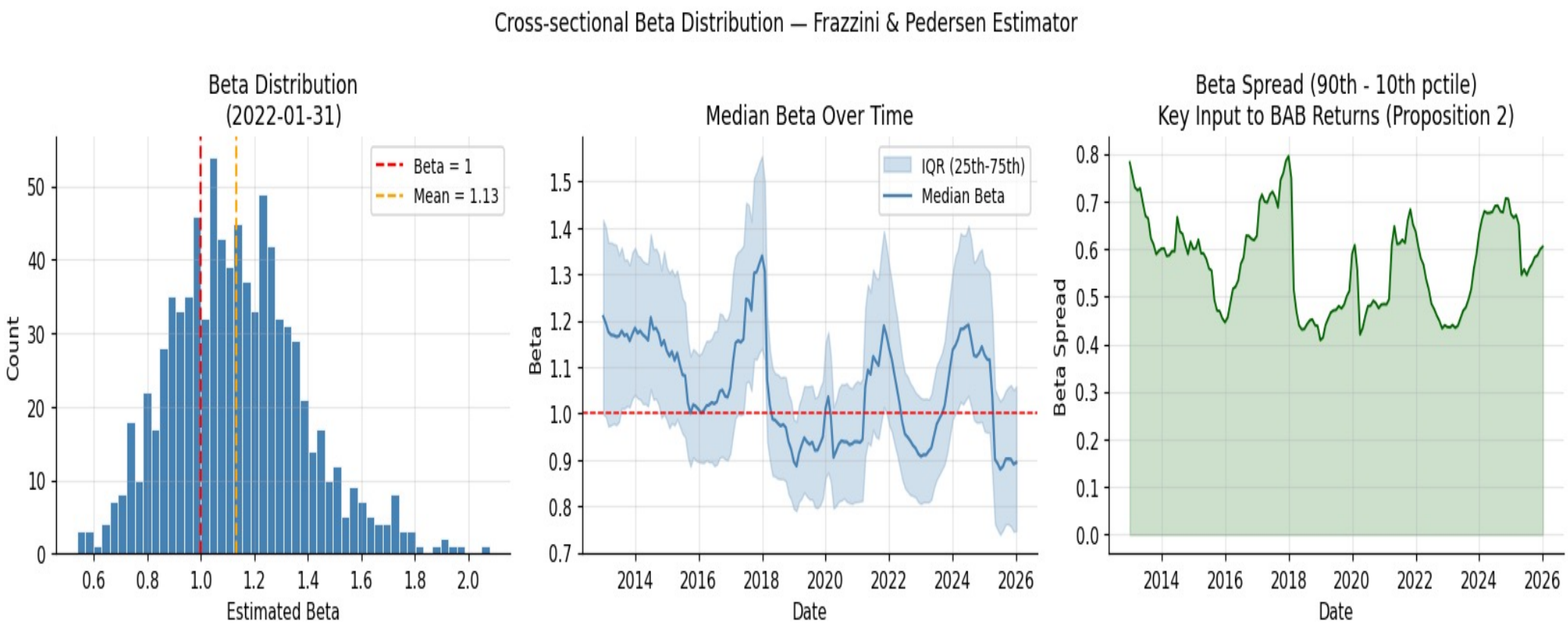
Macro Regimes in Sample

- 2010–2012: Post-GFC recovery, QE1/QE2, low rates.
- 2013–2019: Long bull market, gradual rate normalization.
- 2020: COVID crash and V-shaped recovery (extreme volatility).
- 2021–2022: Inflation surge, rapid Fed tightening.
- 2023–2025: Soft landing, AI-driven tech rally.

Key Limitation: Effective Backtest Period

- With a 5-year correlation window, first valid beta estimates appear around late 2014. Effective backtest period: ~2015–2025 (≈10 years). This is much shorter and covers a different regime than the paper's 1926–2012 sample.

Cross-Sectional Beta Distribution



Left: Beta distribution snapshot (Jan 2022) — bell-shaped, centered near 1 after Vasicek shrinkage. Middle: Median beta and IQR over time — note IQR compression during crises (beta compression). Right: Beta spread (90th–10th pctile) — key input to BAB expected returns per Proposition 2.

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Statistical Validation

Beta-sorted decile portfolios and the empirical Security Market Line

Beta-Sorted Decile Portfolio Construction

Monthly Rebalancing Protocol

- Each calendar month: sort all stocks with valid betas into 10 decile portfolios (D1 = lowest beta, D10 = highest beta), weighted equally within each decile.
- No look-ahead bias: portfolio assignments at end of month t use betas estimated through month t. Returns computed in month t+1.

Step 1 Estimate betas	Step 2 Rank & sort	Step 3 Form portfolios	Step 4 Compute returns	Step 5 Analyze alpha
Compute FP beta for each stock at month-end	Sort 865 stocks by $\hat{\beta}$ into 10 deciles	Equal-weight stocks within each decile	Hold for one month; record portfolio return	OLS regression on market returns

The Core Test (Proposition 1)

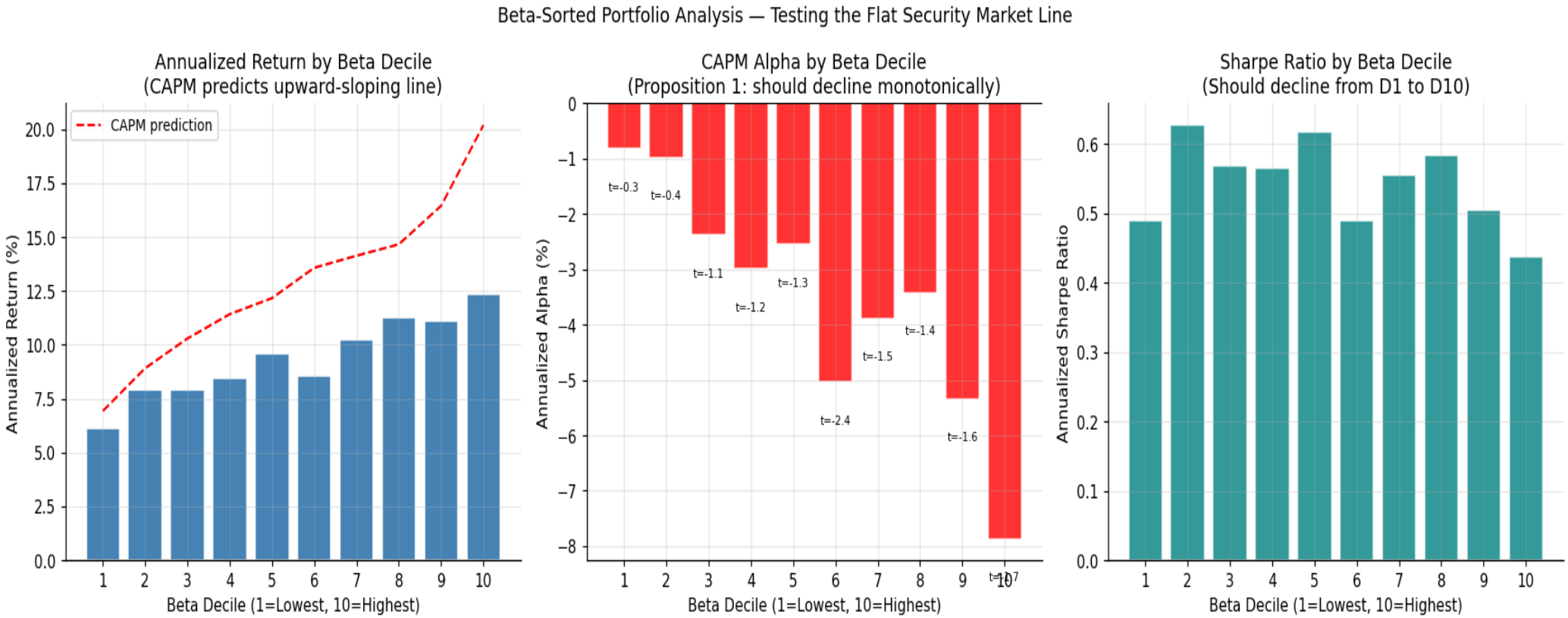
- If BAB holds: returns across deciles should be roughly flat; CAPM alphas should decline monotonically from D1 (positive) to D10 (negative).

Beta-Decile Portfolio Statistics

Portfolio	Ann. Return	Ann. Vol	Sharpe	Beta (realized)	Alpha (ann.)	t(α)
D1 (Lowest β)	**6.1%**	**12.5%**	**0.490**	**0.574**	**−0.8%**	**−0.31**
D2	7.9%	12.7%	0.627	0.739	**−1.0%**	−0.37
D3	7.9%	13.9%	0.569	0.853	**−2.4%**	−1.06
D4	8.4%	14.9%	0.565	0.947	**−3.0%**	−1.18
D5	9.6%	15.6%	0.618	1.008	**−2.6%**	−1.31
D6	8.5%	17.4%	0.490	1.125	**−5.0%**	−2.35
D7	10.2%	18.4%	0.556	1.171	**−3.9%**	−1.46
D8	11.2%	19.3%	0.584	1.215	**−3.2%**	−1.19
D9	11.9%	21.2%	0.561	1.302	**−4.4%**	−1.55
D10 (Highest β)	**10.2%**	**23.1%**	**0.442**	**1.452**	**−10.0%**	**−2.73**

D1 = lowest-beta; D10 = highest-beta. Effective backtest: Jan 2013 – Dec 2025 (156 months). rf = 0 approximation.

Testing the Security Market Line — Visual Evidence



Left: Empirical SML — bars are actual returns, dashed line is CAPM prediction. Bars below the line signal the anomaly. Center: CAPM alphas by decile — declining pattern consistent with Proposition 1; D6 and D10 alphas statistically significant. Right: Sharpe ratios — generally declining toward high-beta deciles.

What Do We See? — Interpreting the SML Results

Paper Findings (1926–2012)

- BAB CAPM alpha: +0.73%/month (FF3 model), t-stat = 7.39.
- 4-factor alpha: +0.55%/month, t-stat = 5.59.
- Sharpe ratios decline monotonically across all 10 deciles.
- Pattern consistent across all four 20-year sub-periods.
- Universe: ~55,600 CRSP stocks over 86 years.

Our Replication (2013–2025)

- SML directionally consistent — empirical SML is flatter than CAPM predicts.
- Alpha pattern broadly declining but noisy in middle deciles.
- D10 alpha: −10.0% annualized; D6 alpha: −5.0% (both negative, as predicted).
- Sharpe ratio pattern mostly declining, with some noise.

Why Our Results Are Noisier — Three Key Reasons

- Short sample: ~120 monthly observations per decile vs. thousands in the paper. High estimation noise in individual decile alphas.
- Large-cap bias: BAB is empirically strongest in small-cap stocks. Our universe of 865 large-cap survivors understates the effect.
- Bull market headwind: 2010–2025 dominated by high-beta growth/tech stocks outperforming — directly opposite to BAB's short leg.

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Constructing the BAB Factor

Rank-weighting, market-neutral scaling, and the portfolio mechanics

BAB Factor: Portfolio Construction Recipe

Rank-Weighting Within Each Group (Equation 16)

$$w_i^L = (\bar{z} - z_i)^+ / \Sigma_j (\bar{z} - z_j)^+ \quad w_i^H = (z_i - \bar{z})^+ / \Sigma_j (z_j - \bar{z})^+$$

Lowest-beta stock gets largest weight in long leg; highest-beta gets largest weight in short leg.

BAB Return — Market-Neutral Scaling (Equation 17)

$$r_{\{t+1\}}^{BAB} = (1/\hat{\beta}_t^L)(r_{\{t+1\}}^L - r^f) - (1/\hat{\beta}_t^H)(r_{\{t+1\}}^H - r^f)$$

Construction Steps

- Step 1: Rank stocks by β̂ at month-end.
- Step 2: Below-median stocks → long leg (L); above-median → short leg (H).
- Step 3: Apply rank-weights — extreme betas get largest weights.
- Step 4: Scale long side by $1/\hat{\beta}^L$ (leverage up to β=1).
- Step 5: Scale short side by $1/\hat{\beta}^H$ (de-leverage to β=1).

Key Properties

- Market-neutral by construction: long and short legs each have β=1 after scaling, netting to zero market beta.
- Self-financing: the short proceeds fund the long position.
- Approximation: rf = 0 throughout (reasonable for 2010–2021 low-rate era).
- Real-world costs NOT modeled: borrowing costs, margin, bid-ask spread.

Realized portfolio betas (our sample): Long side avg $\hat{\beta}^L$ = 0.833 | Short side avg $\hat{\beta}^H$ = 1.354 | Beta spread = 0.520

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Backtesting the BAB Strategy

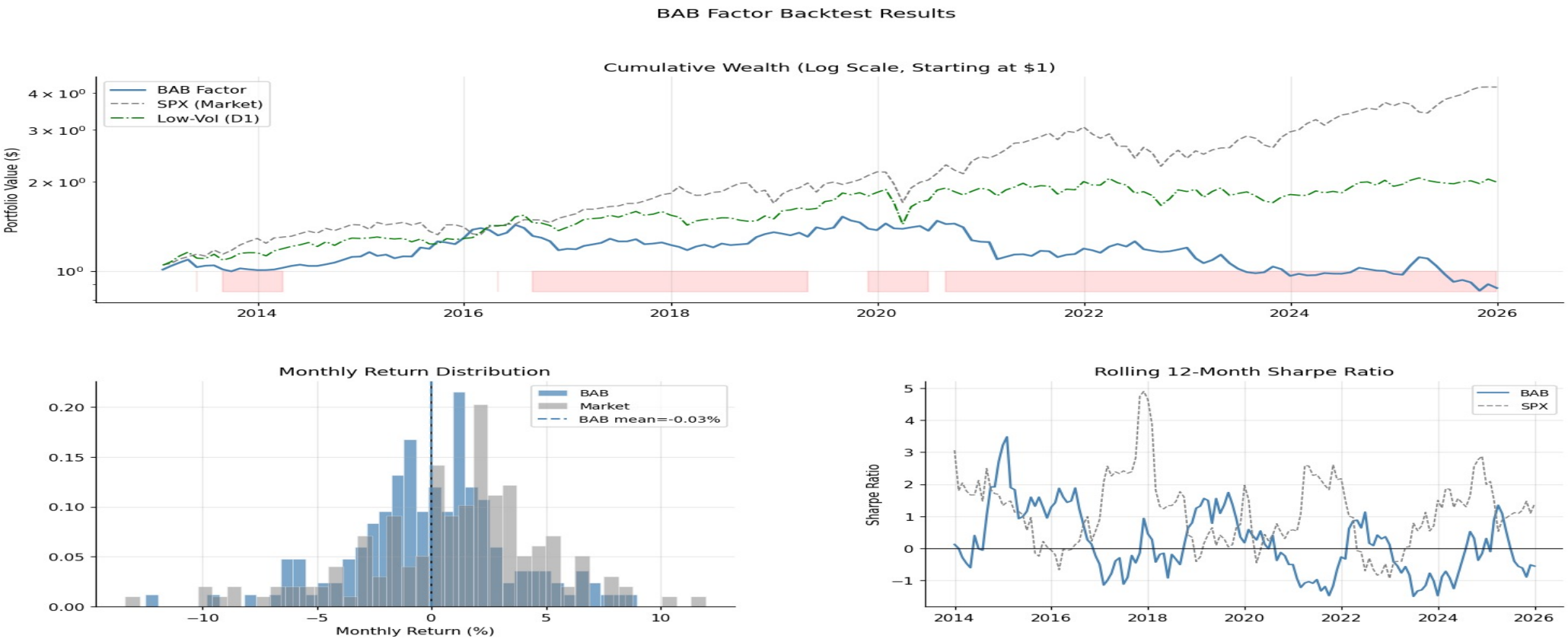
Performance analysis, drawdowns, and comparison to benchmarks

Backtest Performance Summary

−0.025	2.2%	0.72	−43.8%
BAB Sharpe Ratio	BAB CAPM α (ann.)	t-stat on Alpha	Max Drawdown

Metric	BAB Factor	SPX (Market)	Low-Vol D1
Ann. Return	−0.30%	12.07%	6.11%
Ann. Volatility	11.93%	14.17%	12.47%
Sharpe Ratio	**−0.025**	**0.851**	0.490
Max Drawdown	**−43.77%**	−26.33%	−23.70%
CAPM Alpha (ann.)	2.19%	0.00%	−0.82%
t(Alpha)	0.72	1.04	−0.31
Market Beta	−0.206	1.000	0.574
Hit Rate	51.28%	67.95%	57.69%

BAB Factor Performance — Cumulative Wealth & Rolling Sharpe



Top: Cumulative wealth (log scale). Red shading = BAB drawdown > 5%. Bottom-left: Monthly return distribution — inspect mean and tail risk. Bottom-right: Rolling 12-month Sharpe — identifies periods of strategy stress (2020 post-COVID recovery, 2023 AI rally).

Reading the Backtest — Critical Assessment

What the Paper Found (1926–2012)

- BAB annualized Sharpe ratio: 0.78 — highest of all factors studied.
- About 2× the value Sharpe and 40% higher than momentum.
- Market beta of BAB factor: close to 0 by construction.
- Positive returns in all four 20-year sub-periods.
- Average leverage: $1.40 long, $0.70 short.

What Our Sample Shows

- Negative annualized return (−0.30%) over the 2013–2025 period.
- CAPM alpha positive (2.2% ann.) but t-stat only 0.72 — not significant.
- Market beta = −0.206: slight negative market exposure (not perfectly neutral).
- Large max drawdown (−43.8%) reflecting 2020–2021 high-beta recovery.

Statistical Power Perspective

- t-stat scales with √T. Paper's t-stat of 7.4 over 86 years → expected t ≈ 7.4 × √(10/86) ≈ 2.5 with 10 years (if same true alpha).
- Our t-stat of 0.72 is below even that reduced benchmark — suggesting our sample period is particularly unfavorable (tech bull market headwind).
- With ~156 monthly observations, any single decade can easily show the opposite of a long-run average, even for a genuine anomaly. This is NOT a disproof.

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Comparing Our Results to the Paper

Exhibit mapping, metric comparisons, and the BAB Sharpe in context

Paper vs. Replication: Side-by-Side Comparison

Metric	Paper (Frazzini & Pedersen 2014)	Our Replication
Sample period	Jan 1926 – Mar 2012 (86 yr)	~2015–2025 (~10 yr)
Universe	All CRSP stocks (~4,000+)	865 large-cap survivors
BAB Ann. Return	~8–10% (inferred)	−0.3%
BAB Sharpe Ratio	0.78	−0.025
BAB CAPM α (ann.)	~6.5% (0.55%/mo × 12)	2.2%
t-stat on Alpha	5.59 – 7.39	0.72
Market Beta (BAB)	~0 (by construction)	−0.206
Avg long-side beta	~0.70 (leveraged to 1)	0.833
Avg short-side beta	~1.40 (de-lev. to 1)	1.354

Quantitative differences are largely explained by: (1) short sample, (2) large-cap only universe, (3) survivorship bias, (4) tech bull market regime.

The Paper's 0.78 Sharpe in Context

0.78	0.55	0.40	0.40
BAB Factor Sharpe	Momentum (UMD)	Value (HML)	Market (MKT)

Sharpe ratios from Frazzini & Pedersen (2014), U.S. sample 1926–2012. BAB was the highest-Sharpe factor in the study.

Exhibit Mapping: Our Charts vs. Paper Exhibits

- Section 3 SML chart (Panel B) → Figure 1 (declining alphas). Section 3 Sharpe chart → Table III Panel A. Section 5 cumulative return → Figure 3. Section 5 backtest table → Table III (BAB column).

Why BAB Attracted Practitioner Interest

- Theoretically grounded: based on leverage constraint equilibrium, not just data-mining.
- Market-neutral by construction: low correlation to standard factor tilts.
- Multi-asset-class evidence: confirmed across equities, bonds, FX, and commodities.
- Highest Sharpe among all factors in the study's comparison.

Post-Publication Caveats

- Evidence of return decay post-2014 consistent with institutional arbitrage.
- Crowded factor: many managers run BAB-like strategies, compressing the premium.
- Transaction costs: borrowing costs (50–300 bps), monthly rebalancing (~50–100 bps round-trip), reduce live Sharpe by 1–3% annually.

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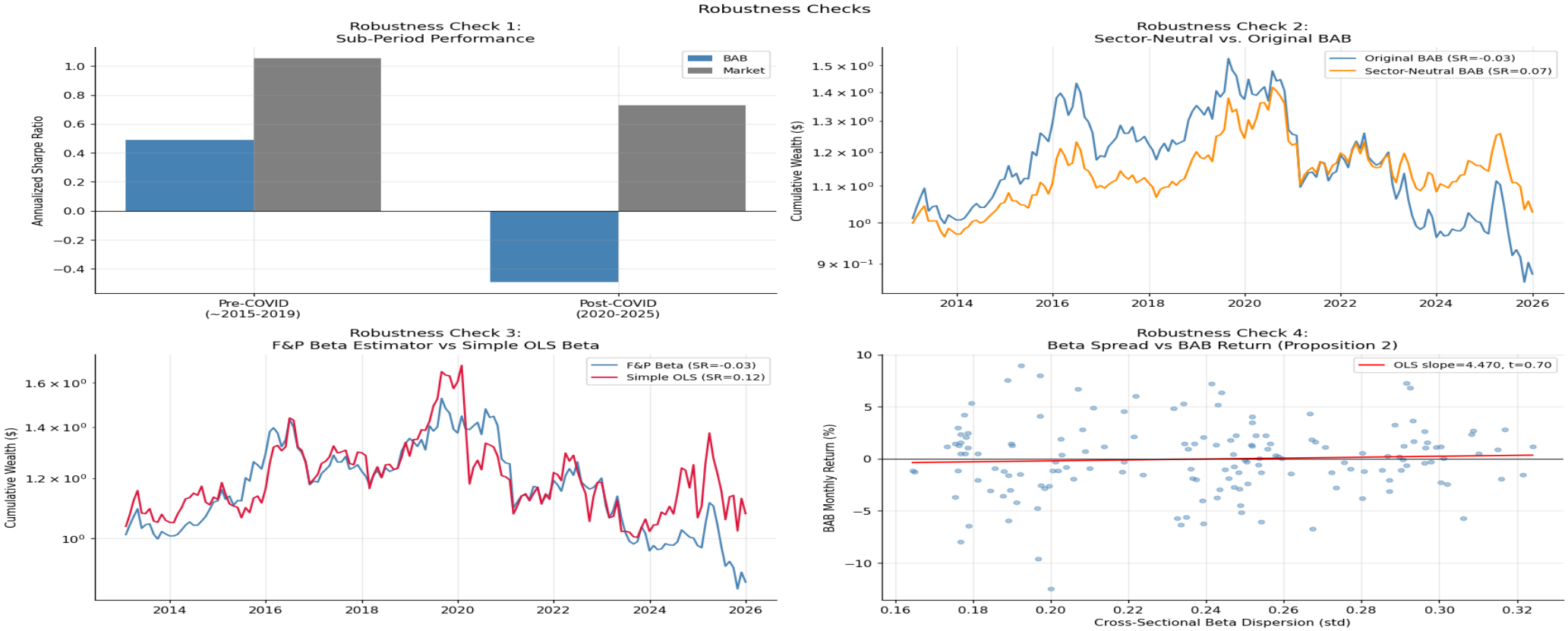
Robustness Checks

Sub-period analysis, sector-neutrality, beta estimation sensitivity

Robustness Checks — Four Dimensions



Top-left: Sub-period Sharpe (pre/post COVID). Top-right: Sector-neutral vs. original BAB — similar behavior confirms beta signal works within sectors. Bottom-left: Simple OLS beta vs. FP estimator. Bottom-right: Proposition 2 test — BAB returns vs. beta spread.

Structural Limitations of This Replication

Data & Sample Limitations

- Sample length: 10 effective years vs. paper's 86 years. t-stat scales with √T: even the same true alpha would yield t ≈ 2.5 vs. paper's 7.4.
- Universe: BAB is strongest in small caps. Our 865 large-cap survivors systematically understate the effect.
- Survivorship bias: all stocks survived 2010–2025; excludes delisted high-beta failures that would have hurt D10.

Regime & Implementation Limitations

- Market regime: 2010–2025 dominated by a tech-led bull market — arguably the worst regime for short high-beta.
- Implementation costs (not modeled): short-selling borrowing (50–300 bps), leverage costs, rebalancing (~50–100 bps round-trip). Reduces Sharpe by est. 1–3% annually.
- rf = 0 approximation: slightly misstates scaled returns in high-rate periods (2022–2023).

What the Literature Says Since 2014

- Pre-2012 replication: multiple academic studies confirm the BAB premium robustly in the original sample.
- Post-publication returns: some evidence of decay post-2014, consistent with institutional arbitrage crowding out the premium.
- International evidence: confirmed across 20 countries in the original paper; subsequent work broadly confirms cross-country validity.
- Time-series dynamics: BAB returns negatively predicted by TED spread (credit conditions) and positively related to beta dispersion — as theory predicts.

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Conclusions & Final Assessment

Key takeaways, the honest verdict, and practical implications

Key Takeaways

1. Leverage constraints flatten the SML: when constrained investors bid up high-beta stocks, low-beta stocks earn higher risk-adjusted returns than CAPM predicts.
2. The FP beta estimator separates volatility (1-year window) from correlation (5-year overlapping window) to reduce noise; Vasicek shrinkage pulls toward the prior of $\beta=1$.
3. Empirical SML test (Proposition 1): our decile portfolios show a directionally flat SML and mostly declining CAPM alphas, consistent with the paper.
4. BAB construction is market-neutral by design: rank-weighting maximizes spread; $1/\beta$ scaling brings both legs to market-beta of 1.
5. Our 10-year backtest returns a negative Sharpe (−0.025) — driven by sample shortness, large-cap bias, survivorship bias, and the tech bull market regime.
6. Statistical power: with 10 years vs. 86, t-stat scales with $\sqrt{(10/86)} \approx 0.34\times$. Insignificant results do not disprove the original finding.
7. BAB remains theoretically grounded and empirically well-supported over the long run, but its live performance post-2014 is weaker due to crowding and transaction costs.

Discussion Questions

1. If leverage constraints are the mechanism behind BAB, what market developments since 2014 might explain the weaker post-publication performance? Which investor types are now less constrained?

2. Proposition 4 predicts BAB returns compress during funding crises (when the TED spread widens). Can you identify episodes in our 2013–2025 sample consistent with this prediction?

3. Our replication uses equal weighting within deciles. The paper uses rank-weighting. How would switching to rank-weighting (as in the BAB factor itself) likely affect the decile portfolio results?

4. The sector-neutral BAB chart shows whether the anomaly operates within or across sectors. What would you conclude if the sector-neutral version underperformed the original? What if it outperformed?

5. An investor claims: 'Your replication shows BAB does not work — negative Sharpe over 10 years.' How would you respond, and what additional evidence would you present to support or refute this claim?

The Honest Verdict

Final Assessment

Frazzini & Pedersen (2014) present one of the most theoretically grounded and empirically well-supported anomalies in the factor investing literature. The intuition is clean (leverage constraints flatten the SML), the construction is rigorous (market-neutral by design), and the evidence spans 86 years and multiple asset classes.

Our 10-year replication in large-cap U.S. stocks during a growth/tech bull market is a partial, noisy window. Directional consistency with the paper's predictions is evidence of robustness; quantitative differences are largely explained by known structural factors.

CAN Conclude from Our Replication

- Empirical SML is flatter than CAPM predicts — consistent with Proposition 1.
- CAPM alphas broadly decline D1→D10 (directionally consistent with the paper).
- BAB construction achieves approximate market-neutrality by design.

CANNOT Conclude from Our Replication

- That BAB produces positive Sharpe in all regimes — our sample is adversarial.
- That the anomaly has disappeared — sample too short, large-cap biased.
- That BAB is directly implementable at low cost — costs not modeled.

For a practitioner, BAB remains credible in a diversified factor portfolio — best accessed through institutional vehicles with managed costs.

References & Further Reading

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Questions & Discussion